Enformer

Effective gene expression prediction from sequence by integrating long-range interactions

Avsec, Agarwal, Visentin, Ledsam, Grabska-Barwińska, Taylor, Assael, Jumper, Kohli & Kelley (2021)

Nature Methods

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CS781 – AI for Health Sciences

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📖 Paper Overview

Avsec, Ž., Agarwal, V., Visentin, D., Ledsam, J. R., Grabska-Barwińska, A., Taylor, K. R., Assael, Y., Jumper, J., Kohli, P., & Kelley, D. R. (2021).
Effective gene expression prediction from sequence by integrating long-range interactions.
Nature Methods, 18, 1196–1203.
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One-sentence summary: A sequence-to-function model that uses self-attention after a convolutional stem to reach a ~100 kb receptive field, improving multitask prediction of thousands of human and mouse genomic tracks and downstream variant-effect benchmarks.

Scale (paper, Methods): 196,608 bp input; **5,313 human** + **1,643 mouse** tracks at **128 bp** resolution; trained on **64 TPU v3** cores (~**150,000** steps, ~**3** days).

🎯 Objectives

1. **Background** – Gene regulation, distal enhancers, and why long-range context matters
2. **Key questions** – What CNN-based models miss; what Enformer changes
3. **Algorithm** – Convolutional tower + transformer + organism-specific heads; receptive field to ~100 kb
4. **Implementation** – Data, loss, training hardware, train/val/test split
5. **Results** – Expression tracks, enhancer prioritization, eQTL variant effects, MPRA / CAGI5
6. **Limits & critique** – What the paper claims, what remains open, what not to overclaim clinically

🧬 Why this problem matters

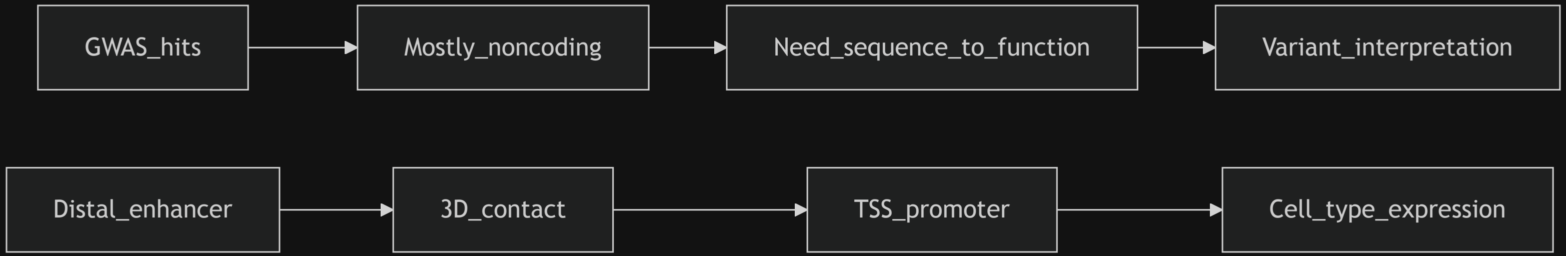
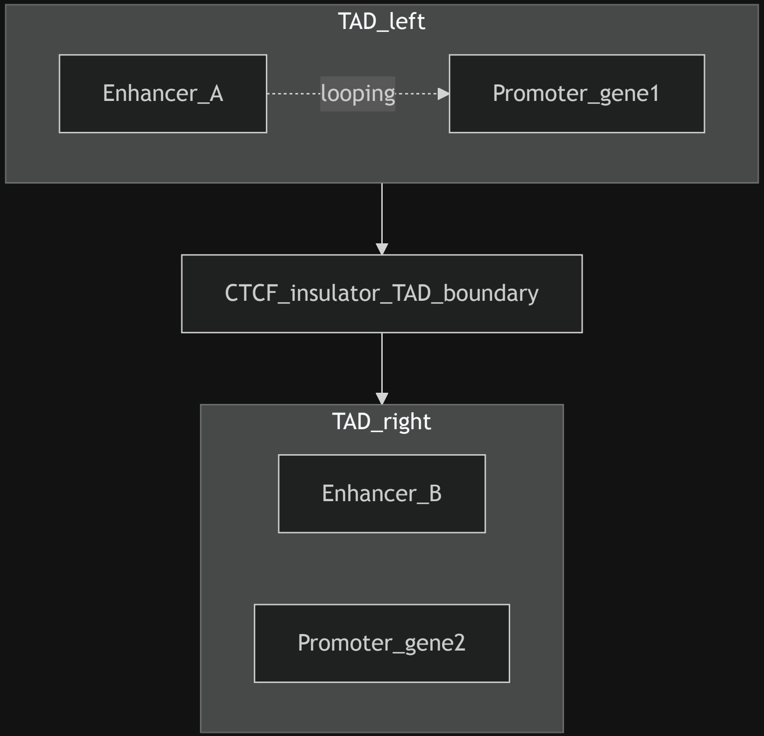


Diagram: my schematic of the motivating problem (not a paper figure).

- **Noncoding variants** dominate GWAS; mechanisms run through **cis-regulation** (enhancers, promoters, insulators).
- **Experimental validation** does not scale to every variant in every tissue.
- **Computational models** that map **DNA → regulatory readouts** are a complement to population genetics, not a replacement.

🧭 Background: enhancers, promoters, and TADs



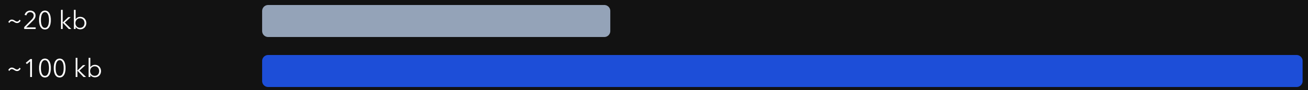
- **Promoter** – region near the **TSS (transcription start site)** where the basal transcription machinery assembles.
- **Enhancer** – regulatory DNA that can lie **tens to hundreds of kb away** on the linear genome yet contact the promoter through **3D folding**.
- **TAD (topologically associating domain)** – megabase-scale compartment; **CTCF-rich boundaries** and **insulators** reduce illegitimate enhancer-promoter crosstalk **across** domains (paper Fig. 2c,d; Discussion).

Diagram: teaching schematic (not a paper figure). Paper uses TAD boundary analysis in Fig. 2.

⛔ Key limitation of prior CNN models

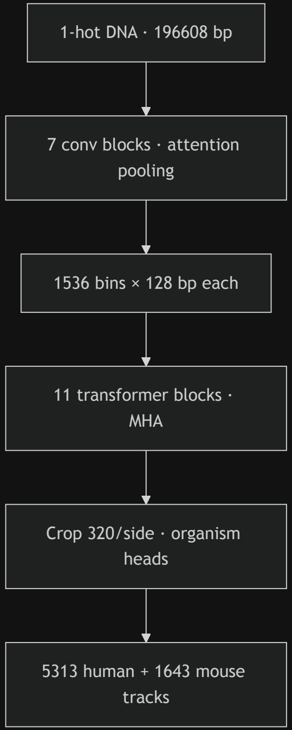
Receptive field (informal): how far along the sequence a prediction at the TSS can effectively "see."

Model family (paper)	~Reach from TSS	High-confidence enhancer-gene pairs "in view" (paper estimate)
Basenji2 / ExPecto	**~20 kb**	**47%** within 20 kb
Enformer	**~100 kb**	**84%** within 100 kb



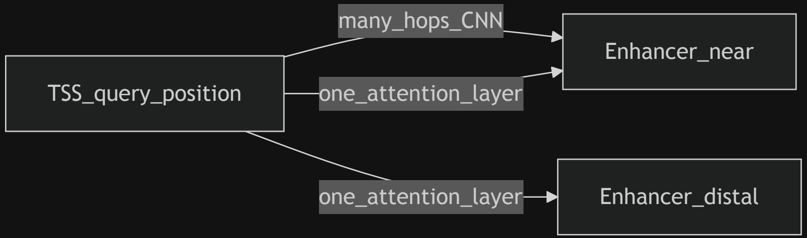
Percentages: Introduction / Results narrative in Avsec et al. comparing enhancer-gene distances to model reach.

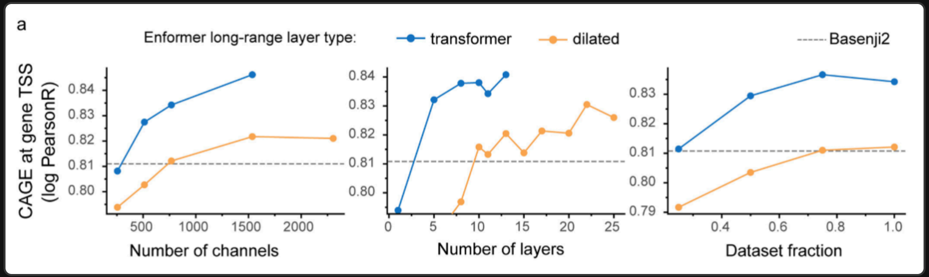
🏗️ Enformer architecture (overview)



- **Paper fact:** 7 conv blocks, 11 transformer blocks, then organism heads (Fig. 1a; Extended Data Fig. 1).
- **Output window: 896** bins × **128 bp** = **114,688 bp** of predicted tracks along the sequence center (Methods).

🔍 Why self-attention here (vs. dilated CNNs)





Top: intuition sketch. Bottom: Extended Data Fig. 5a (validation curves).

- **Convolution:** local mixing; reaching distal positions needs **many stacked layers** (dilated or not).
- **Self-attention:** each position forms a **weighted sum over all positions**; one layer can link **TSS ↔ distal enhancer** directly (paper Results + Ext. Data Fig. 5a).
- **Relative positional encodings:** custom basis functions (not plain NLP sin/cos) so the model can encode **distance and strand direction** to the TSS (Ext. Data Fig. 6; Methods).

🧪 Implementation: data, loss, and training

Topic	Detail (paper, Methods)
Targets	**5,313** human tracks (TF ChIP, histone ChIP, DNase/ATAC, CAGE) + **1,643** mouse tracks
Loss	**Poisson negative log-likelihood** per position vs observed counts (same as Basenji2)
Train / val / test	**34,021** train, **2,213** val, **1,937** test sequences; **homology-aware** split via syntenic **1 Mb** regions (human–mouse bipartite graph)
Optimization	**Adam** lr **5×10^{-4}**, **5k**-step warmup, global grad clip **0.2**, **150k** steps, batch **64** on **64** TPU v3 cores (~**3** days); alternate human/mouse batches
Augmentation	Random shift up to **3 bp** + **reverse complement** (matches Basenji2)
Fine-tune	Extra **30k** steps on human at lr **1×10^{-4}** (paper Methods)

📈 Results: gene expression (CAGE at TSS)

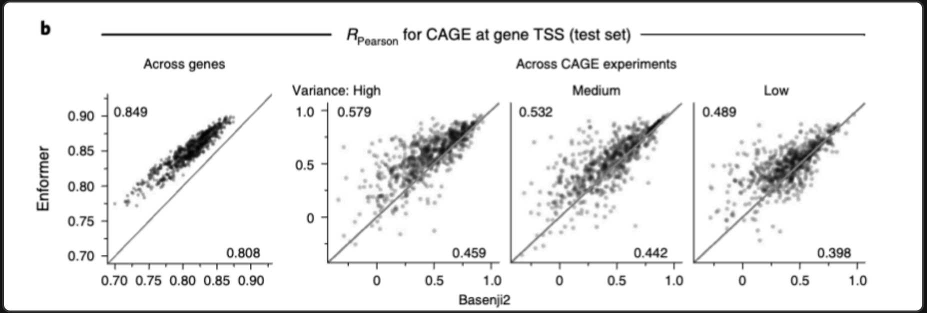


Fig. 1b: Pearson correlation of CAGE at TSS (log(1+x), standardized) – across genes per experiment (left) and across experiments per gene (right). Corner means from the paper figure.

Paper headline: mean CAGE correlation **0.81 → 0.85** across genes; improvement **~2×** the Basenji1→Basenji2 jump; closes **~⅓** of the gap to an **~0.94** replicate-level ceiling (Ext. Data Fig. 2; narrative in Results).

Statistical test: paired **Wilcoxon**, **$P < 10^{-38}$** vs Basenji2 on Fig. 1b/c-style comparisons (paper).

> **Framing:** strong held-out performance on gene bodies and TSS readouts; still not an experimental replicate.

📊 Results: all four assay classes (human test)

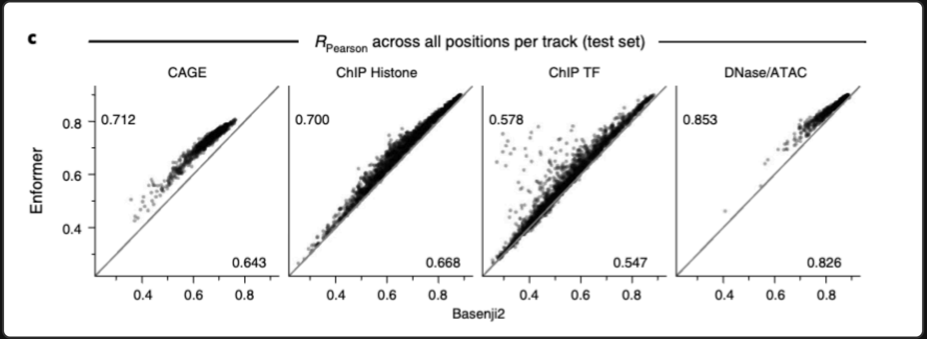


*Fig. 1c: each dot is one of **5,313** tracks – Pearson correlation across all 128-bp bins on held-out human sequence (paper).*

- **CAGE, ChIP histone, ChIP TF, DNase/ATAC** – Enformer **>** **Basenji2** for every dot cloud (paired **Wilcoxon**, **$P < 10^{-38}$**).
- **Interpretation:** the receptive-field upgrade helps **more than just one assay type**; largest gains emphasized for **CAGE** (distal regulation; paper Discussion).

Results: enhancer prioritization (CRISPRi)

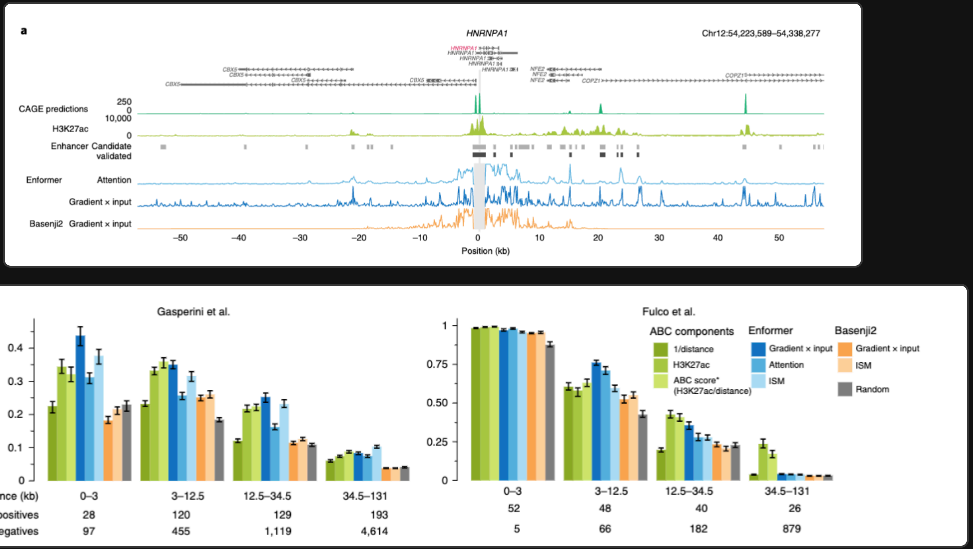


*Fig. 2a-b: contribution scores (gradient×input, attention, ISM) vs CRISPRi-validated enhancer-gene pairs (Gasperini; Fulco). Enformer **DNA-only** scores vs **ABC** (often uses Hi-C + H3K27ac; paper).*

🏛 Results: TAD boundaries and insulator-like behavior

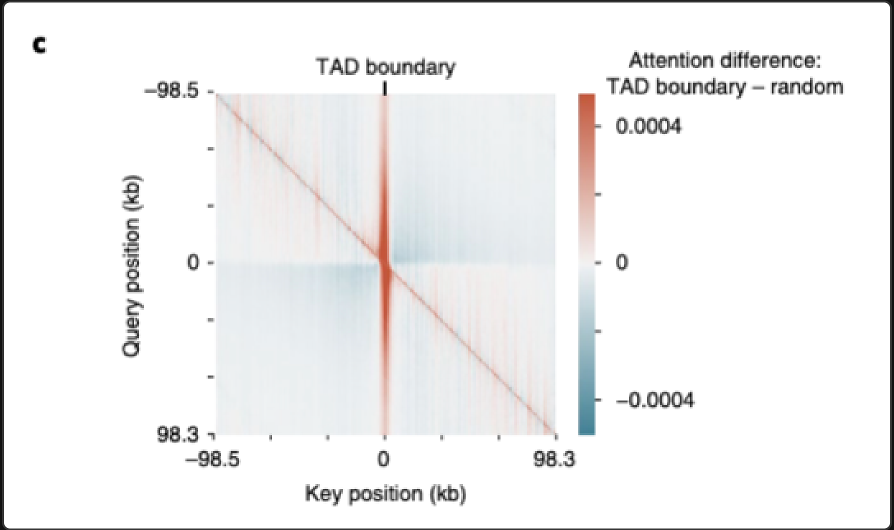
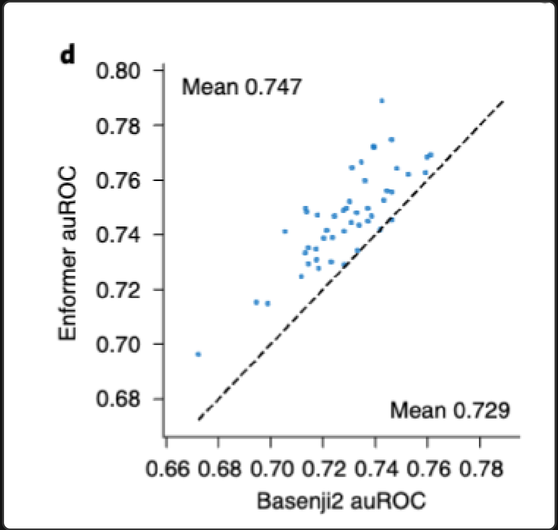
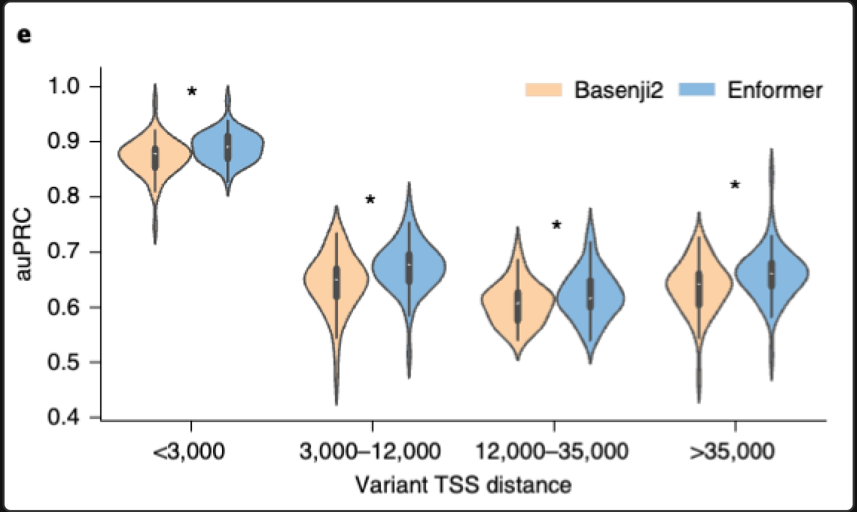


Fig. 2c (heatmap): interpretability evidence from sequence, not a clinical claim.

- **On this slide – Fig. 2c:** average attention-matrix **difference** for **1,500** sequences centered on **TAD boundaries** vs matched random controls – **more** attention on the **boundary stripe**; **less** attention **across** the boundary (**blue** off-diagonal blocks vs random centering).
- **Also in the paper (not shown here): Fig. 2d** quantile plots (**across-TAD** down, **at-boundary** up; Mann–Whitney); **Ext. Data Fig. 8** – **CTCF**-like motifs from gradient×input at boundaries (**unsupervised**).

Results: eQTLs, SLDP, and fine-mapped causal variants





On this slide: *Fig. 3d (left), Fig. 3e (right).* ***Fig. 3a-c*** *in the paper report **SLDP** (genome-wide, LD-aware concordance with GTEx) – summarized in the bullets, not reproduced here.*

- **Fine-mapped classification (Fig. 3d; SuSiE PIP):** random forests on **Δprediction** features – **47 / 48** tissues favor Enformer; mean **auROC 0.729 → 0.747**.
- **Distance bins (Fig. 3e):** gains at **all** TSS-distance bins (paired Wilcoxon **P < 10^{-4}**).

Results: MPRA saturation mutagenesis (CAGI5)

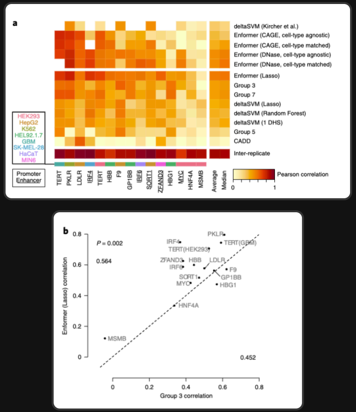


Fig. 4: 15 disease-linked loci, CAGI5 train/test splits; compare Enformer (lasso on 5,313 Δfeatures, or training-free PC summary) vs competing methods (paper).

- **Headline:** Enformer lasso best average correlation across loci vs seven other method families; beats CAGI5 winning Group 3 (paired one-sided Mann–Whitney, P = 0.002, Fig. 4b).
- **Training-free scores** from Enformer still competitive – strong inductive bias from multitask sequence training (paper Discussion).

🌟 Strengths (what the paper actually contributes)

Contribution	Evidence (paper)	Why it matters
Long-range receptive field ~100 kb	vs ~20 kb Basenji2/ExPecto; ablations Ext. Data Fig. 5	Captures most high-confidence enhancer–gene pairs in their estimate
Multitask sequence → tracks	5,313 human + 1,643 mouse outputs; Poisson training	One model shares statistical strength across assays
Enhancer prioritization from DNA	Fig. 2 vs CRISPRi + ABC comparisons	Prioritization without Hi-C at prediction time
Variant-effect benchmarks	GTEx fine-mapping + MPRA / CAGI5	Bridges population genetics + functional assays

⚠️ Limitations and improvements

[SOURCE: paper] – authors highlight

- Only **cell types / assays present in training**; **no zero-shot** to unseen tissues or marks without new data.
- **3D contacts** (Hi-C, etc.) are **not** explicit inputs; artful fusion with sequence models like **Akita / DeepC** could help (Discussion cites Fudenberg, Schwessinger).
- **Resolution vs compute:** 128 bp bins trade off against quadratic attention cost; finer bins need more memory (Discussion).
- **Variant sensitivity** could improve with **more CRISPR / MPRA** used as training signal, not only evaluation (Discussion).

[Context beyond the paper – my critique]

- **Correlation ≠ mechanism:** better tracks do not automatically yield causal narratives in patients.
- **Tissue matching is hard:** SLDP and eQTL analyses required manual CAGE–GTEx matching; mismatches dilute signal (Methods acknowledge skipping ambiguous cases).
- **Sign prediction at distal sites:** Ext. Data Fig. 10 shows **both models struggle** when variants are far from promoters – honest frontier.
- **Deployment:** compute, calibration, prospective validation, and equity across ancestries remain outside this paper's scope.

🔬 Translational framing + course bridge

Paper stance: improved fine-mapping and variant prioritization are **scientifically plausible** next steps; authors release **precomputed** 1000 Genomes scores to lower the engineering barrier.

[Context beyond the paper – Anton]

- **Clinical genomics** still needs **prospective validation**, appropriate **intended-use** scoping, and **laboratory follow-up** – Enformer is not a standalone diagnostic.
- **Course bridge (CS781 Paper Ideas memo):** the first anchor talk was **EHR representation learning**; Enformer is the **sequence-to-function** anchor. Connecting **AoU (All of Us)** respiratory phenotypes to **cis-regulatory** hypotheses is a natural place for models like Enformer for **prioritizing noncoding variants** – but **not** a substitute for cohort design, ancestry-aware validation, and wet-lab confirmation.

🎯 Key takeaways

For the paper (facts + claims they support)	Takeaway (how to read it)
~100 kb receptive field improves **multitask** prediction vs **Basenji2**	Ask **which enhancers** enter the window for your variant of interest
Attention + custom relative PEs beat **dilated CNN** ablations in their sweeps	It is **architecture + inductive bias**, not a transformer trophy
Strong enhancer prioritization + **GTEx / MPRA** benchmarks	Benchmarks test **specific pipelines**; **LD / tissue matching** matter
Released weights + variant scores	Useful **research tool**; **not** automatic clinical truth

📚 References

Primary

- Avsec, Ž., Agarwal, V., Visentin, D., Ledsam, J. R., Grabska-Barwińska, A., Taylor, K. R., Assael, Y., Jumper, J., Kohli, P., & Kelley, D. R. (2021). Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18, 1196–1203. https://doi.org/10.1038/s41592-021-01252-x

Key dependencies cited in talk (selection)

- Kelley, D. R. et al. Basenji / Basenji2 (cross-species regulatory prediction). *PLoS Comput. Biol. / Genome Res.* (as in Enformer references).
- Gasperini, M. et al.; Fulco, C. P. et al. CRISPRi enhancer maps (Fig. 2).
- GTEx Consortium; Wang et al. SuSiE fine-mapping resource (Fig. 3).
- Kircher et al. saturation mutagenesis; Shigaki et al. CAGI5 (Fig. 4).

🙏 Thank you